Quantum Junction Challenge: Condensed Codex Session Report

Codex run summary

Built 2026-06-06 08:18:38

Executive Summary

The session produced selected peak bitstrings for **41/49** challenge files. The remaining blanks are `104_49, 48_42, 56_43, 64_44, 72_45, 80_46, 88_47, 96_48`. The selected set contains **10** exact statevector answers and **31** tensor-network or MPS-derived candidates. Among the approximate selected rows: **17** from quimb GPU; **11** from quimb CPU; **1** from Aer MPS; **1** from MPO unswap; **1** from RCM CPU. Exact statevector peaks had mean peak probability 0.513 and mean gap to the second bitstring 0.459.

The main result is a usable candidate set, not a proof of correctness for every approximate row. Easy and moderate instances often had large sampled peak fractions and/or corroborating evidence. Several hard rows are weak: they were selected because they were the best available evidence, but their sampled top fractions are near one or two samples out of 512/1024, and the newly selected MPO-unswap row for `64_40` is similarly diffuse. The session later pivoted away from long-running graph-MPS sweeps after the user set a practical "about 2 hours" per-attempt budget.

Candidate Selection Rule

The collector selected one answer per challenge using source priority, then evidence consistency. The priority order used in the current collector was: exact statevector, canonical Quimb GPU, optimized-U3 GPU, canonical Quimb CPU, MPO-unswap, RCM/MST/ degree/mid/fast/identity CPU fallbacks, sparse beam, and low-priority Aer MPS pilot evidence. Exact answers were accepted directly. Approximate answers were chosen as sampled top bitstrings in Qiskit/counts order, with the right-most bit corresponding to qubit 0. When multiple methods agreed on the same bitstring, the report records that as higher evidence count.

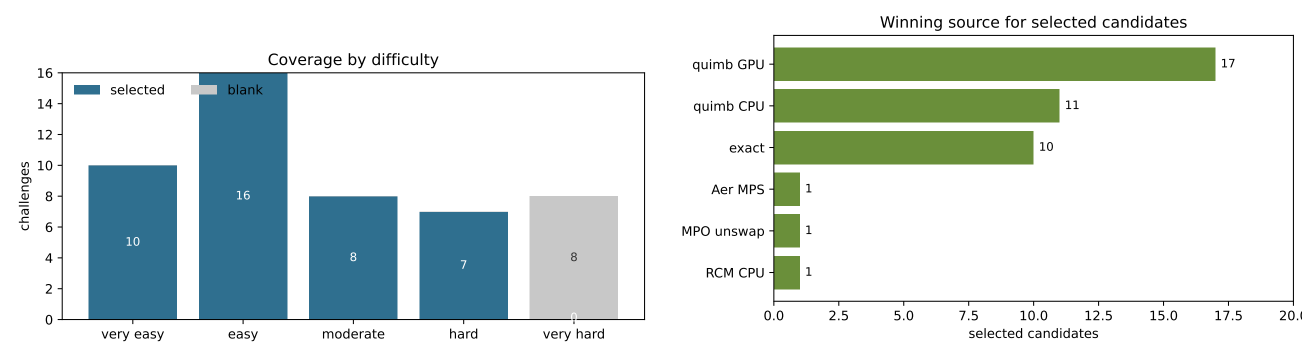

Figure 1: Coverage and winning evidence sources. The 8 missing rows are all very hard.

What Was Tried

Method	Purpose and implementation	Outcome

Exact statevector	Used Aer/statevector on small enough circuits to establish ground-truth peaks and bit ordering.	10 exact answers. Used as calibration and highest-priority evidence.
Canonical graph/tree MPS	Ran Quimb tree-ordered MPS sampling on CPU and GPU, using sampled top bitstrings as candidate peaks.	Main source of approximate answers; strong on many easy/moderate rows and weak but usable on some hard rows.
Ordering fallbacks	Tried RCM, MST, degree, mid, fast, and identity orderings to change contraction behavior.	Added corroboration; RCM supplied the selected 56_38 weak candidate. Long jobs were later cancelled.
Sparse computational-basis beam	Added a C++ top-K sparse branch propagator through RX/RZ/CX.	Rejected for answer selection: on known 16_28 the exact answer was only rank 3.
Static QASM forensics	Counted gates, angle classes, pair repetitions, connectivity, leading one-qubit prefixes, and operation 5-gram similarity.	Confirmed all circuits are RX/RZ/CX/SWAP style, very-hard files are a distinct dense-obfuscation regime, and Clifford shortcuts are not appropriate.
Optimized-U3 transpilation	Transpiled the remaining blanks to U3+CX to reduce single-qubit gate count.	Reduced total gate count by roughly 39–42% on blanks; not yet selected as evidence.
MPO unswapping	Patched the local Kremer-Dupuis style MPO cancellation/unswapping code: parameterized SABRE trials, fixed Quimb local expectation API, fixed quantum-op progress counting.	Calibrated on known 16_28 in 93.26 s and recovered the exact answer. The completed 64_40 run is included as weak evidence; the remaining very-hard runs were still active or incomplete at report build time.

Static Structure

The QASM files use only `rx`, `rz`, `cx`, and sparse `swap` gates. Very-hard instances are much larger and more internally similar to each other than to the easier sets. They also have dense leading single-qubit prefixes and high all-to-all pair coverage. This drove the pivot toward structure-aware MPO cancellation rather than only waiting for generic graph-MPS contractions.

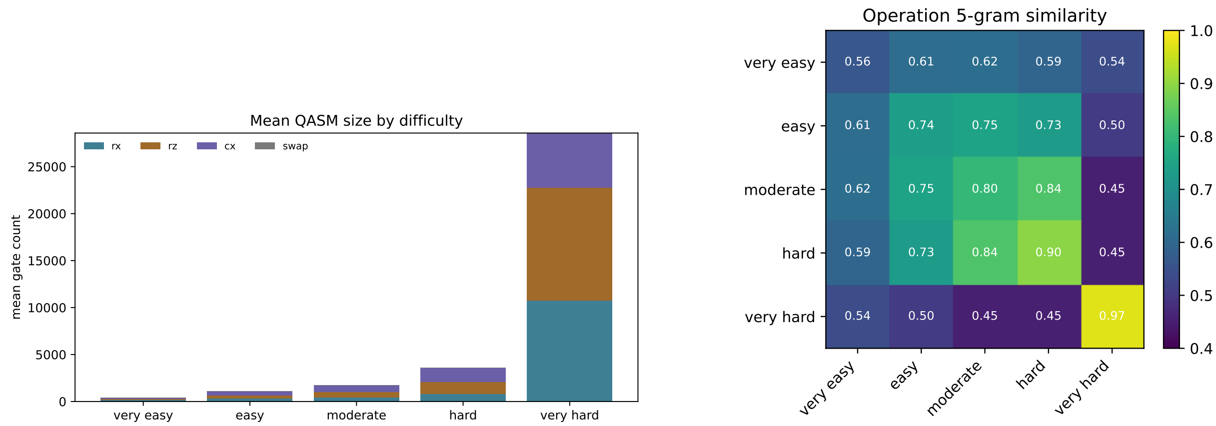

Figure 2: Static QASM summary. Very-hard files are large and operation-sequence-similar to each other.

Exact Baseline

The exact rows were used both as final answers and as calibration targets. The peak is visibly separated from the runner-up on these cases, which made them useful for validating bit order and method behavior.

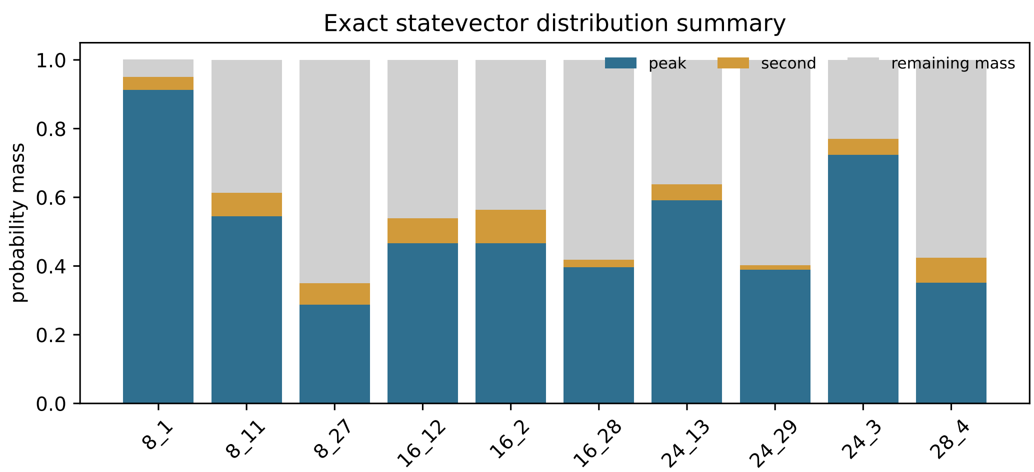

Figure 3: Exact statevector probability mass: peak, second-largest bitstring, and remaining mass.

Challenge	q	exact peak	peak prob.	second prob.	gap
8_1	8	10101101	0.9126	0.0379	0.8747
8_11	8	01001110	0.5454	0.0687	0.4768
8_27	8	11001001	0.2879	0.0627	0.2252
16_12	16	1111000101101011	0.4665	0.0730	0.3936
16_2	16	1010101011001000	0.4670	0.0976	0.3694
16_28	16	1101001111011100	0.3962	0.0231	0.3731
24_13	24	111110011111001011010001	0.5912	0.0472	0.5441
24_29	24	110100010111100001001001	0.3901	0.0134	0.3767
24_3	24	011110010000101010001000	0.7240	0.0471	0.6769
28_4	28	1111111000101010110110011111	0.3511	0.0735	0.2776

Approximate Tensor-Network Results

For approximate rows, the sampled top fraction was the main confidence signal. Strong rows have a clear heavy bitstring. Weak hard rows are retained because they are currently the best available candidates, but they should be interpreted as low-confidence until corroborated.

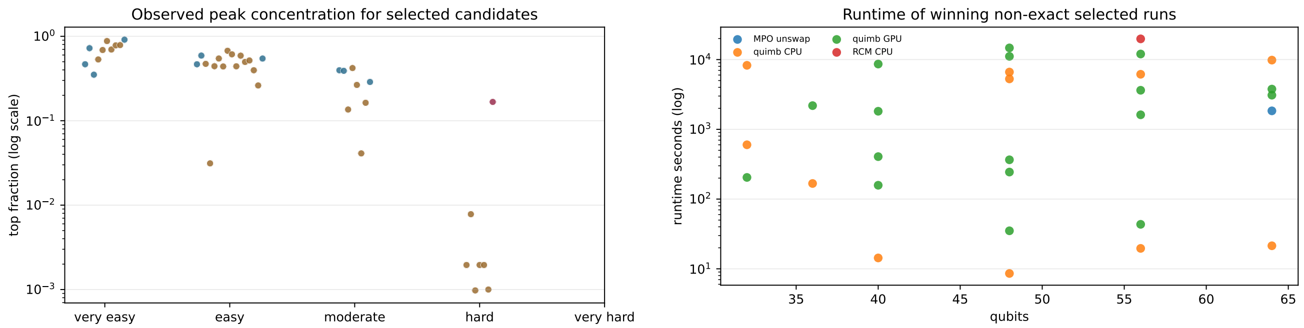

Figure 4: Peak concentration and runtime for selected rows. The hard set contains several low-concentration candidates.

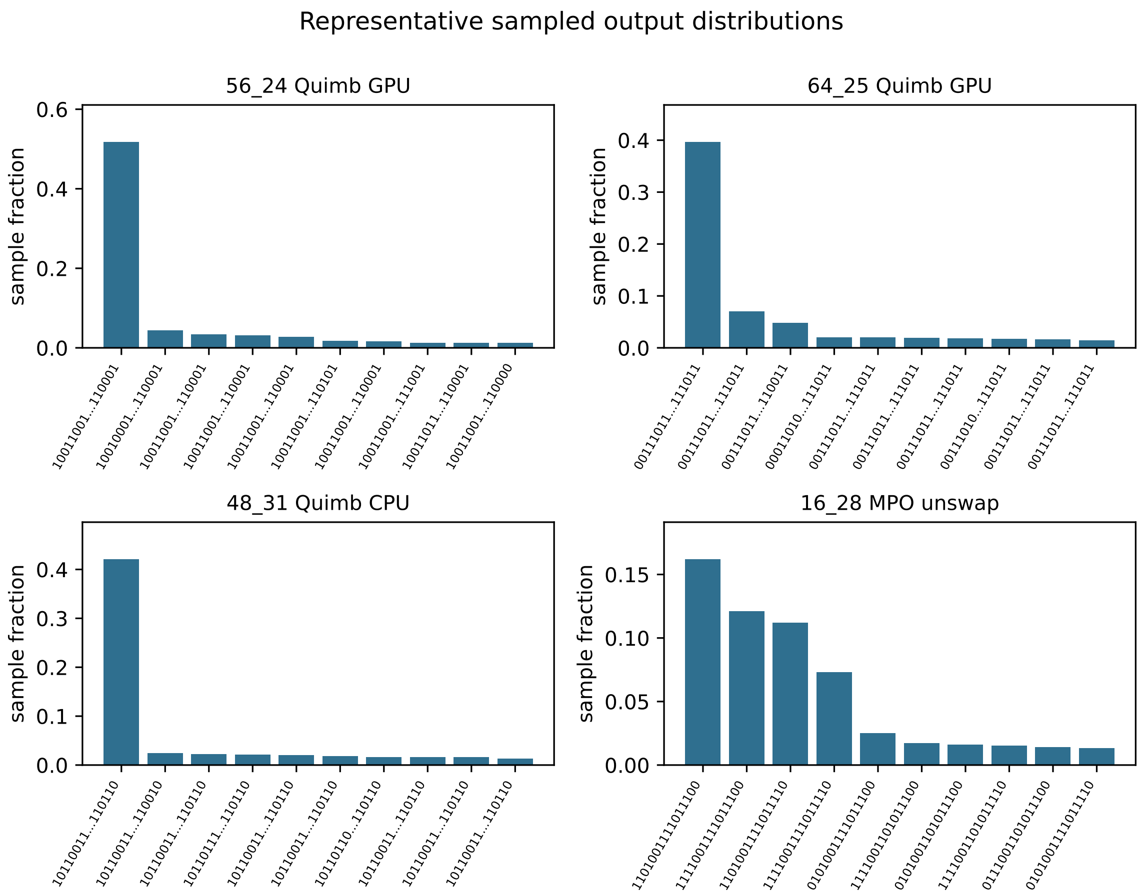

Figure 5: Representative sampled distributions. The first three are selected Quimb candidates; the fourth is the MPO-unswap validation on 16_28.

Weak Selected Rows

Challenge	difficulty	source	top frac.	evidence	candidate
40_35	hard	quimb GPU	0.001953	3	0101100110101111111100001010100111000100 1
48_36	hard	quimb GPU	0.007812	3	001111011011001100110011001000101001101010111000
48_37	hard	quimb GPU	0.0009766	3	100101001010001101010100100101001001000101010001
56_38	hard	RCM CPU	0.001953	1	100110010111100000000100110101011000010010011101 10110001
56_39	hard	quimb GPU	0.001953	3	100101110011010101011111110101101101101100001010 01010000
64_40	hard	MPO unswap	0.001	1	101101001110001011010000000011000011100111101010 0100110100001100
64_41	hard	Aer MPS	0.167	1	000100011101001110011001001111011100000010001101 0001101010110010
32_5	very easy	quimb CPU	0.5322	1	00111000101010100001000000010000
36_6	very easy	quimb CPU	0.6885	1	100110100111101001001101110011001000
40_7	very easy	quimb CPU	0.8789	1	0110111011010001010011111110010011000111
48_8	very easy	quimb CPU	0.6973	1	000100100110111001001111111100101011001110010100
56_9	very easy	quimb CPU	0.7812	1	100101101011001011101001100101100111011001100011 01010100
64_10	very easy	quimb CPU	0.7871	1	001101001011000111001011100110010010110001011111 0110010100101011

Rejected or Not-Yet-Selected Paths

The sparse beam search was attractive because it was fast, but its pruning bias failed calibration. In known 16_28, the exact answer was rank 3 rather than rank 1, so no sparse-beam guesses were promoted to selected answers. U3 optimization reduced the remaining blank gate counts substantially, but its GPU pass was not completed before this report snapshot.

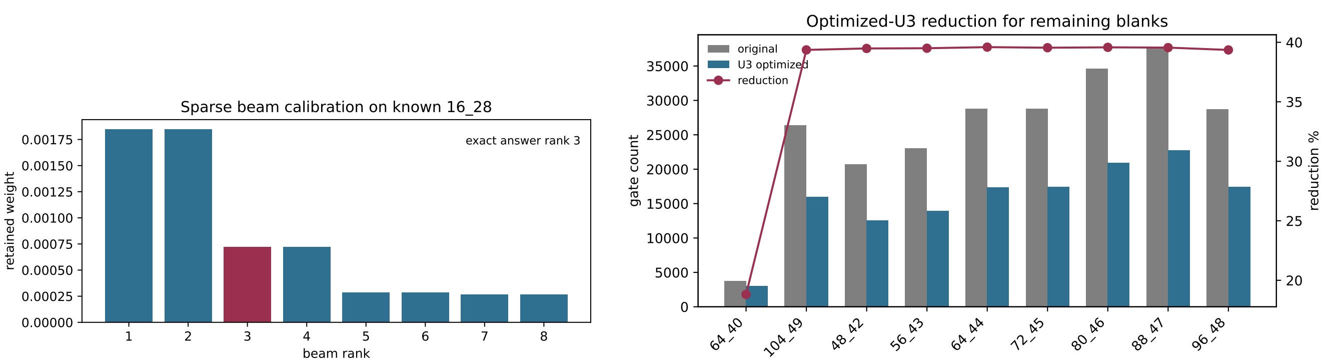

Figure 6: Left: sparse beam failed calibration. Right: U3 optimization reduced the remaining blank circuits but was not yet selected as evidence.

Blank	q	original gates	U3 gates	reduction
64_40	64	3758	3051	18.8%
104_49	104	26361	15985	39.4%
48_42	48	20733	12547	39.5%
56_43	56	23004	13917	39.5%
64_44	64	28787	17389	39.6%
72_45	72	28777	17397	39.5%

80_46	80	34586	20898	39.6%
88_47	88	37660	22764	39.6%
96_48	96	28703	17406	39.4%

Current Slurm Snapshot

The report is an as-of snapshot. Any active unresolved MPO-unswap jobs below were still running or pending when the PDF was built, and their outputs are not counted in the **41/49** result.

JobID	state	elapsed	CPUs	name	node/reason
34614305	RUNNING	1:45:09	4	bash	erc-hpc-comp025
34614255	PENDING	0:00	2	peaked_unswap_rollup	(Dependency)
34614254_[47-48%5]	PENDING	0:00	16	peaked_unswap_unresolved	(JobArrayTaskLimit)
34614254_46	RUNNING	1:19:04	16	peaked_unswap_unresolved	erc-hpc-comp192
34614254_45	RUNNING	1:48:43	16	peaked_unswap_unresolved	erc-hpc-comp202
34614254_41	RUNNING	1:50:15	16	peaked_unswap_unresolved	erc-hpc-comp195
34614254_43	RUNNING	1:50:15	16	peaked_unswap_unresolved	erc-hpc-vm064
34614254_44	RUNNING	1:50:15	16	peaked_unswap_unresolved	erc-hpc-vm065

Method Summary

Source	selected winners	candidate evidence rows	attempted evidence rows
Aer MPS	1	6	6
exact	10	10	10
MPO unswap	1	2	2
quimb CPU	11	36	36
degree CPU	0	1	1
fast CPU	0	5	5
quimb GPU	17	21	21
RCM CPU	1	16	16
sparse beam	0	5	5

Selected Candidate Table

Full candidate bitstrings are shown in wrapped monospace chunks. Blank rows remain unsolved.

Challenge	difficulty	q	candidate	source	validation	top frac.	evidence
8_1	very easy	8	10101101	exact	exact	0.9126	2
16_2	very easy	16	1010101011001000	exact	exact	0.467	2
24_3	very easy	24	011110010000101010001000	exact	exact	0.724	2
28_4	very easy	28	1111111000101010110110011111	exact	exact	0.3511	2
32_5	very easy	32	00111000101010100001000000010000	quimb CPU	unknown	0.5322	1
36_6	very easy	36	100110100111101001001101110011001000	quimb CPU	unknown	0.6885	1
40_7	very easy	40	0110111011010001010011111110010011000111	quimb CPU	unknown	0.8789	1
48_8	very easy	48	000100100110111001001111111100101011001110010100	quimb CPU	unknown	0.6973	1
56_9	very easy	56	100101101011001011101001100101100111011001100011 01010100	quimb CPU	unknown	0.7812	1
64_10	very easy	64	001101001011000111001011100110010010110001011111 0110010100101011	quimb CPU	unknown	0.7871	1
8_11	easy	8	01001110	exact	exact	0.5454	4
16_12	easy	16	1111000101101011	exact	exact	0.4665	4
24_13	easy	24	111110011111001011010001	exact	exact	0.5912	3

32_14	easy	32	00000101001101000001011111101100	quimb GPU	unknown	0.4717	3
36_15	easy	36	110110011011111111011001011110101111	quimb GPU	unknown	0.03125	3
40_16	easy	40	0101110101001110011000111011100110010110	quimb GPU	unknown	0.4424	4
40_17	easy	40	0010010101010111001001000010001000001001	quimb GPU	unknown	0.5459	4
40_18	easy	40	0100000110010010001101111000111111001110	quimb GPU	unknown	0.4395	3
48_19	easy	48	011001010111101100111110000001011101001110010000	quimb GPU	unknown	0.6738	2
48_20	easy	48	1010101001010010100001101010000001011110010000000	quimb GPU	unknown	0.6123	3
48_21	easy	48	111010101110101011110101000001101000100000001001	quimb GPU	unknown	0.4414	3
56_22	easy	56	111001001001100100111100101101100110000001000101 11111011	quimb GPU	unknown	0.5898	2
56_23	easy	56	010011010011111111000001010011101110111000111111 00001101	quimb GPU	unknown	0.4971	2
56_24	easy	56	100110010011111011111110111010111011010100110010 11110001	quimb GPU	unknown	0.5176	4
64_25	easy	64	001110111111000011011011110101100001000001010011 0111000001111011	quimb GPU	unknown	0.3965	4
64_26	easy	64	011010101010001101011101100001110001011011011001 1100011001100110	quimb GPU	unknown	0.2617	3
8_27	moderate	8	11001001	exact	exact	0.2879	2
16_28	moderate	16	1101001111011100	exact	exact	0.3962	6
24_29	moderate	24	110100010111100001001001	exact	exact	0.3901	4
32_30	moderate	32	10011000010011110111101111010111	quimb CPU	unknown	0.1357	2
48_31	moderate	48	101100111000101011111111101010111011011000110110	quimb CPU	unknown	0.4209	2
48_32	moderate	48	011101010111011110001110010101010101101110001110	quimb CPU	unknown	0.2656	2
56_33	moderate	56	110010011001010011110101001000100101011111110111 01010000	quimb CPU	unknown	0.04102	2
64_34	moderate	64	001101010001001100111010111010010100101100101101 1001111011100110	quimb CPU	unknown	0.1631	2
40_35	hard	40	01011001101011111111000010101001110001001	quimb GPU	unknown	0.001953	3
48_36	hard	48	001111011011001100110011001000101001101010111000	quimb GPU	unknown	0.007812	3
48_37	hard	48	100101001010001101010100100101001001000101010001	quimb GPU	unknown	0.0009766	3
56_38	hard	56	100110010111100000000100110101011000010010011101 10110001	RCM CPU	unknown	0.001953	1
56_39	hard	56	100101110011010101011111110101101101101100001010 01010000	quimb GPU	unknown	0.001953	3
64_40	hard	64	101101001110001011010000000011000011100111101010 0100110100001100	MPO unswap	unknown	0.001	1
64_41	hard	64	000100011101001110011001001111011100000010001101 0001101010110010	Aer MPS	unstable_top1_vs_aggregate	0.167	1
48_42	very hard	48					0
56_43	very hard	56					0
64_44	very hard	64					0
72_45	very hard	72					0
80_46	very hard	80					0
88_47	very hard	88					0
96_48	very hard	96					0
104_49	very hard	104					0

Conclusions

The practical output of the session is a 41-row candidate answer set and a clear split between reliable and weak evidence. The exact and high-top-fraction Quimb rows are the most defensible. The hard weak rows should be treated as provisional. The best next direction is not another unconstrained graph-MPS sweep: it is bounded MPO-unswapping or optimized-U3 GPU runs with calibration checks and immediate collector refreshes.

Appendix: Per-Problem Result Distributions

This appendix shows the distribution evidence used for each challenge in this session, ordered by challenge number. For exact rows, the bars are exact statevector probabilities for the top outcomes. For Quimb graph-MPS, MPO-unswapping, and Aer pilot rows, the bars are sampled output fractions from the selected or current evidence source. All bitstrings are in Qiskit/counts order, with the right-most bit corresponding to qubit 0. Panels with no bars had no completed distribution evidence at the report snapshot; those rows were still blank or only had started/pending jobs.

The appendix is deliberately compact: it shows the top reported outcomes rather than every sampled bitstring, because several hard and very-hard runs produced hundreds or thousands of singleton samples. The unlisted fraction annotation is therefore part of the interpretation. A large unlisted fraction means the observed distribution was diffuse and the selected bitstring should be treated as weak evidence.

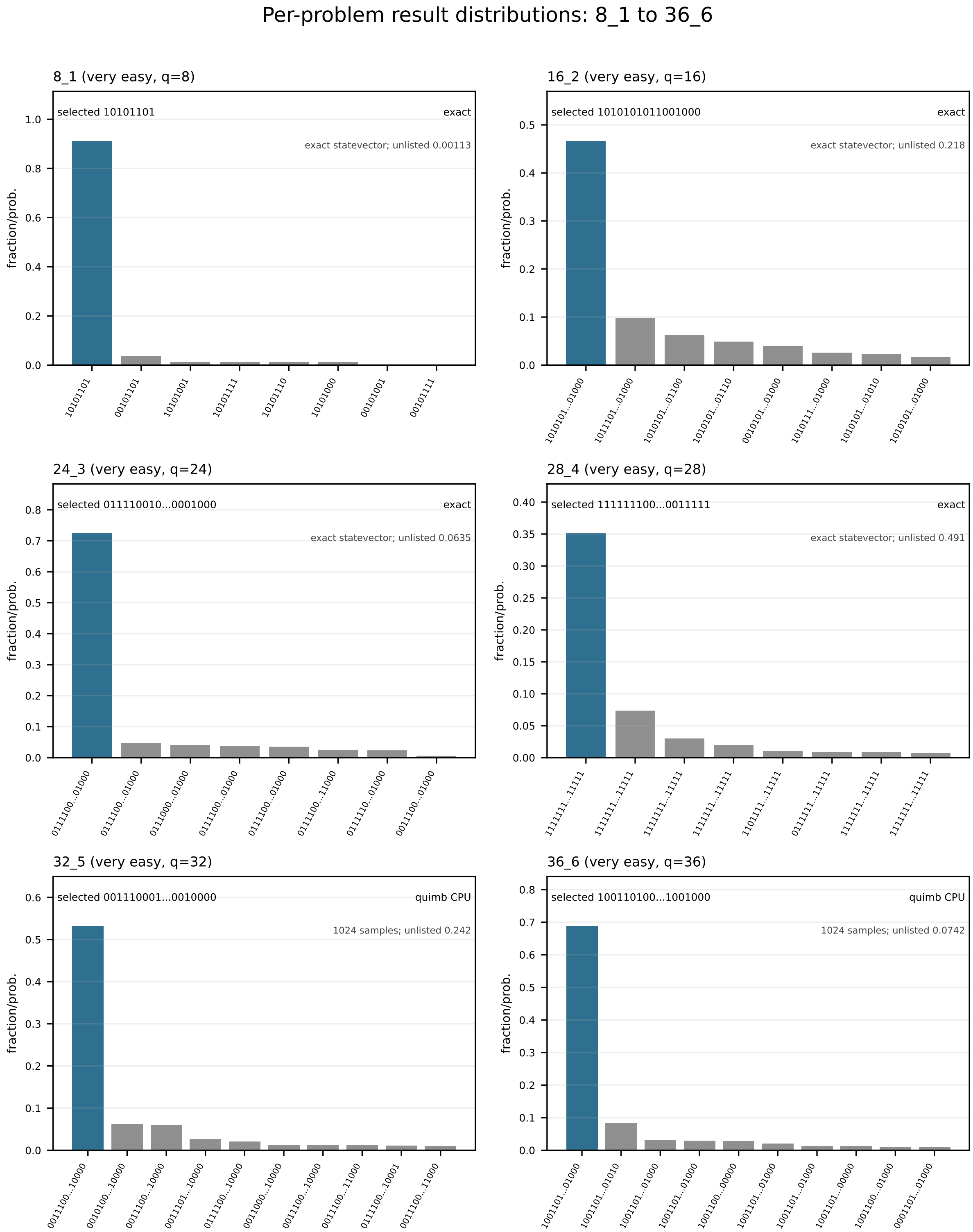

Figure 7: Per-problem result distributions, page 1. Bars show the top reported exact probabilities or sample fractions. The selected candidate is highlighted in blue when it appears among the displayed top outcomes; grey bars are alternate outcomes. Unlisted probability mass or sample fraction is reported inside each panel.

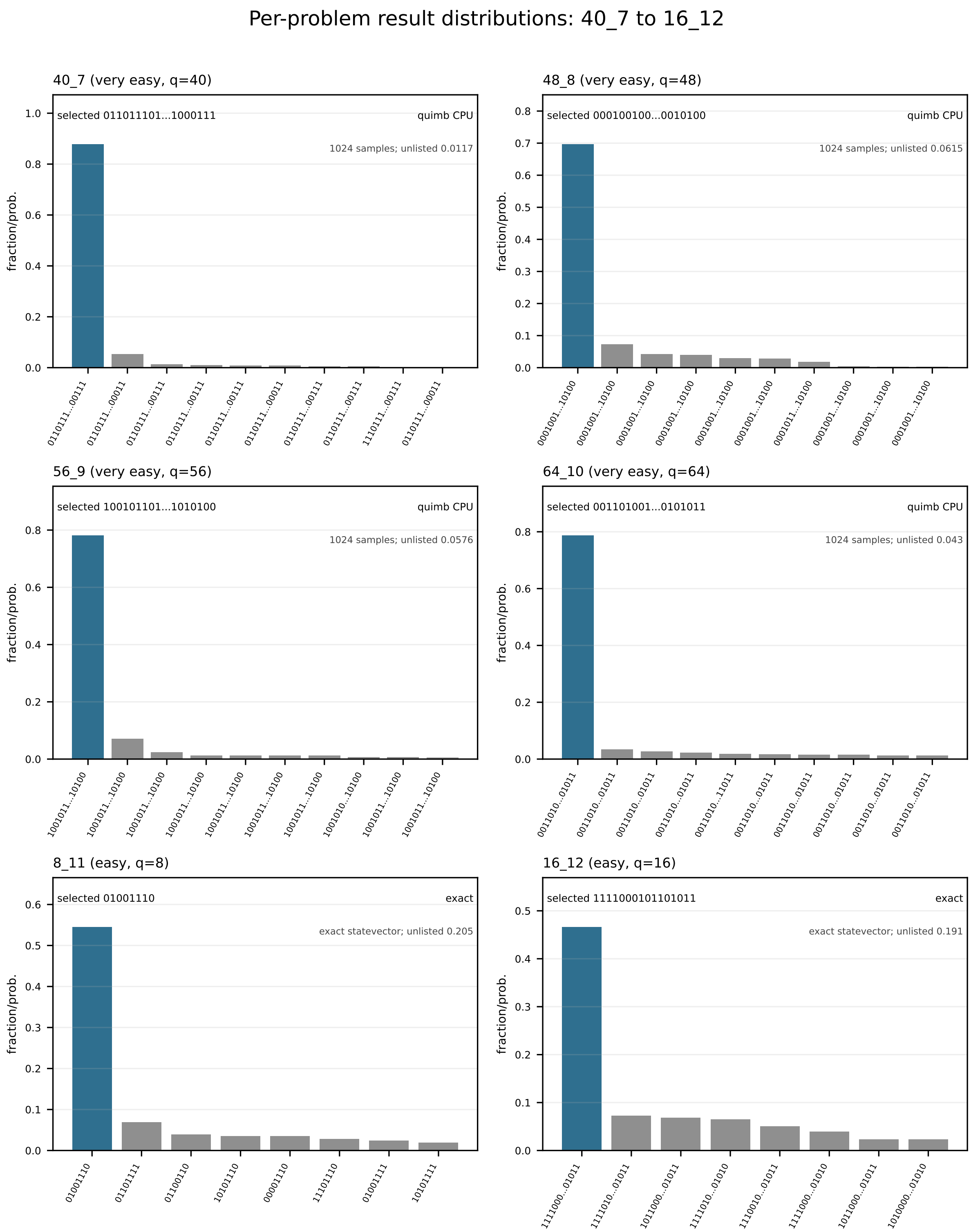

Figure 8: Per-problem result distributions, page 2. Bars show the top reported exact probabilities or sample fractions. The selected candidate is highlighted in blue when it appears among the displayed top outcomes; grey bars are alternate outcomes. Unlisted probability mass or sample fraction is reported inside each panel.

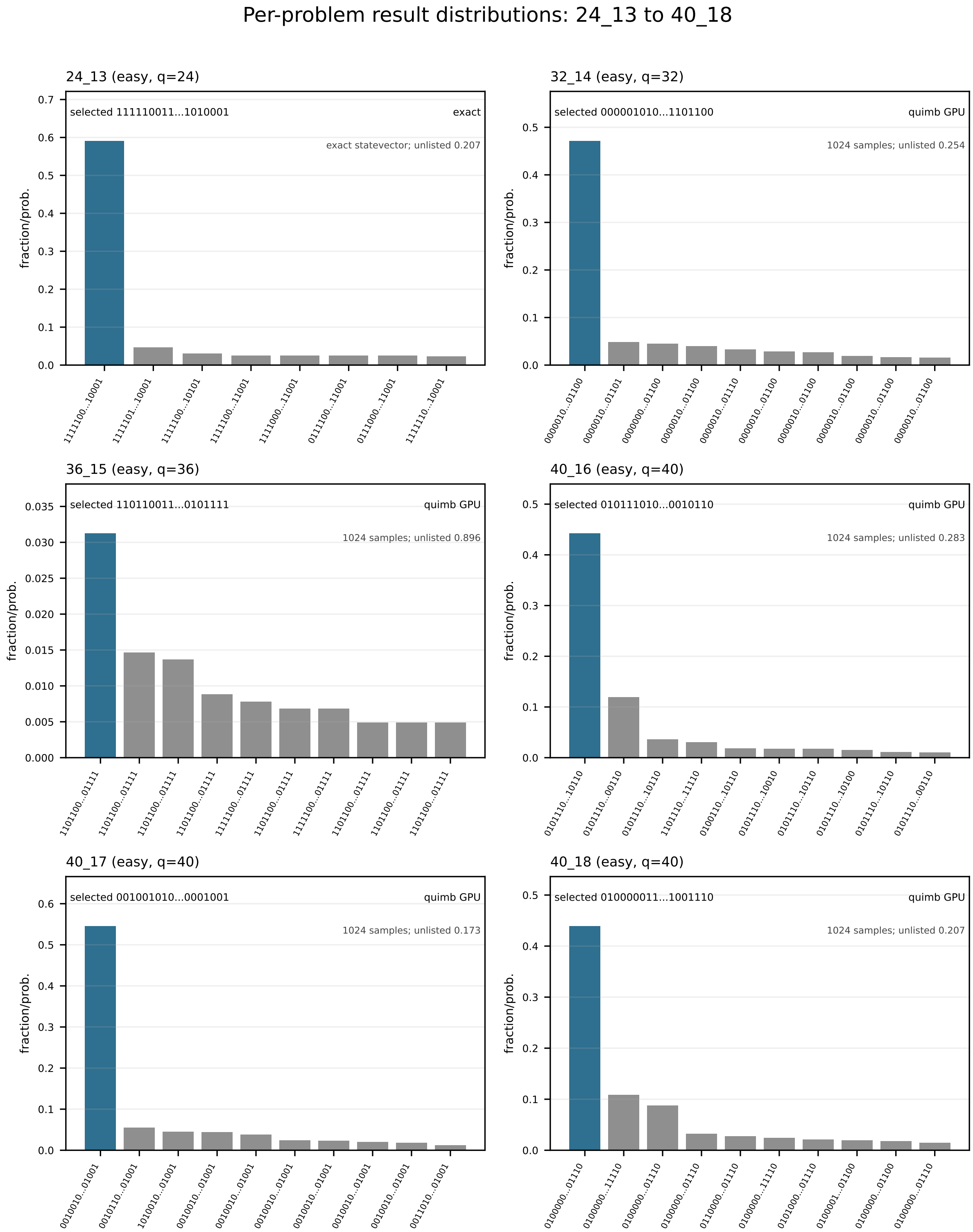

Figure 9: Per-problem result distributions, page 3. Bars show the top reported exact probabilities or sample fractions. The selected candidate is highlighted in blue when it appears among the displayed top outcomes; grey bars are alternate outcomes. Unlisted probability mass or sample fraction is reported inside each panel.

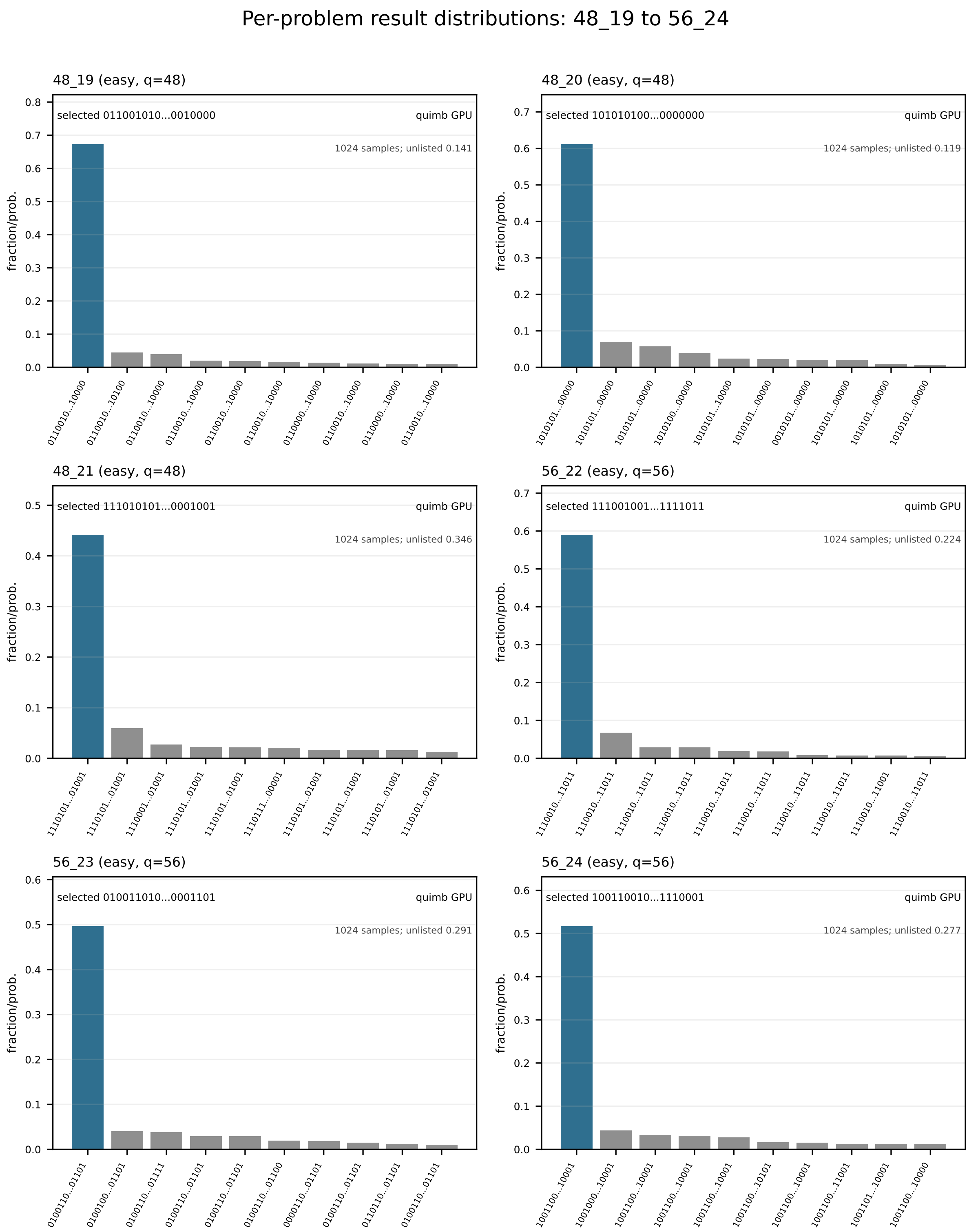

Figure 10: Per-problem result distributions, page 4. Bars show the top reported exact probabilities or sample fractions. The selected candidate is highlighted in blue when it appears among the displayed top outcomes; grey bars are alternate outcomes. Unlisted probability mass or sample fraction is reported inside each panel.

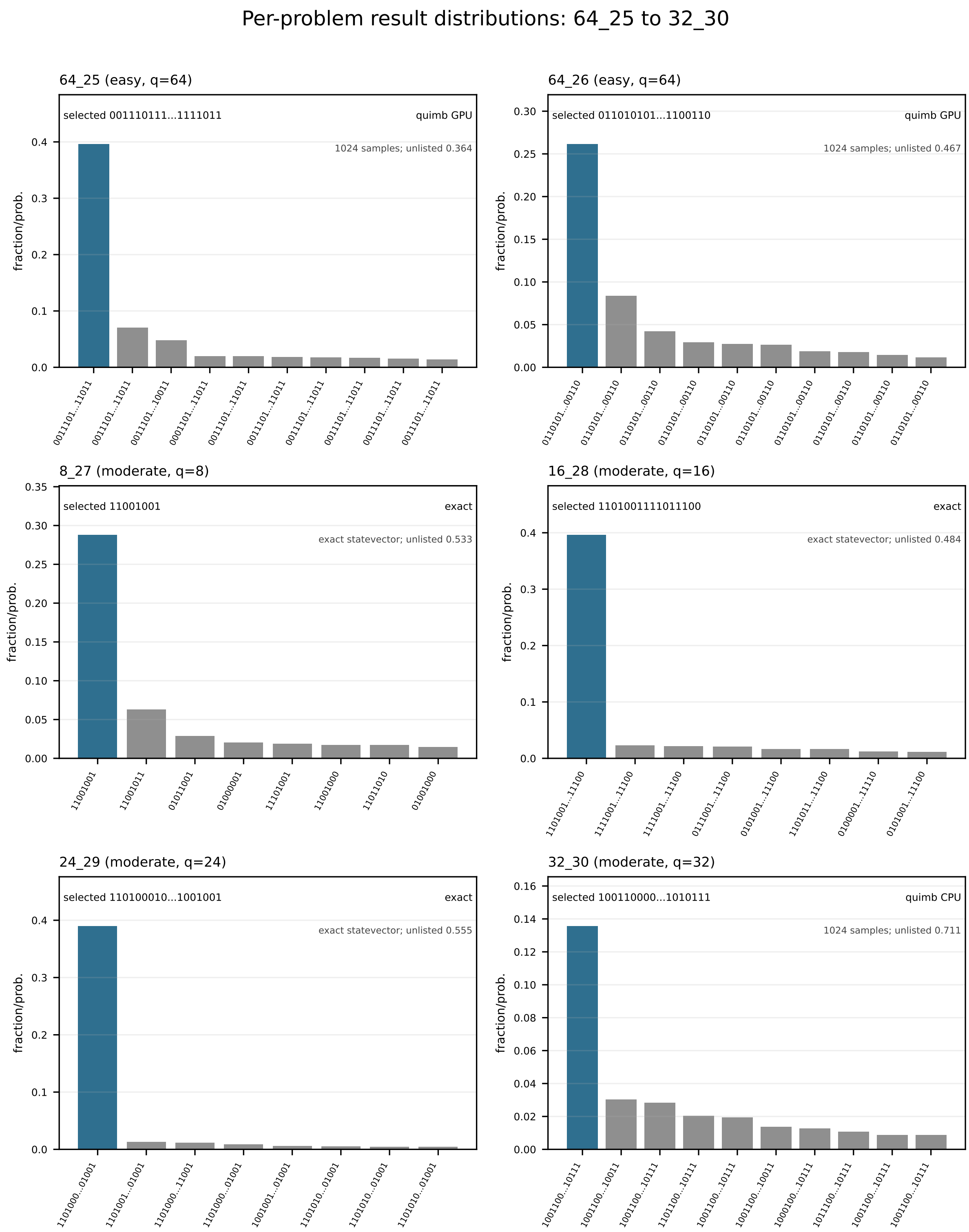

Figure 11: Per-problem result distributions, page 5. Bars show the top reported exact probabilities or sample fractions. The selected candidate is highlighted in blue when it appears among the displayed top outcomes; grey bars are alternate outcomes. Unlisted probability mass or sample fraction is reported inside each panel.

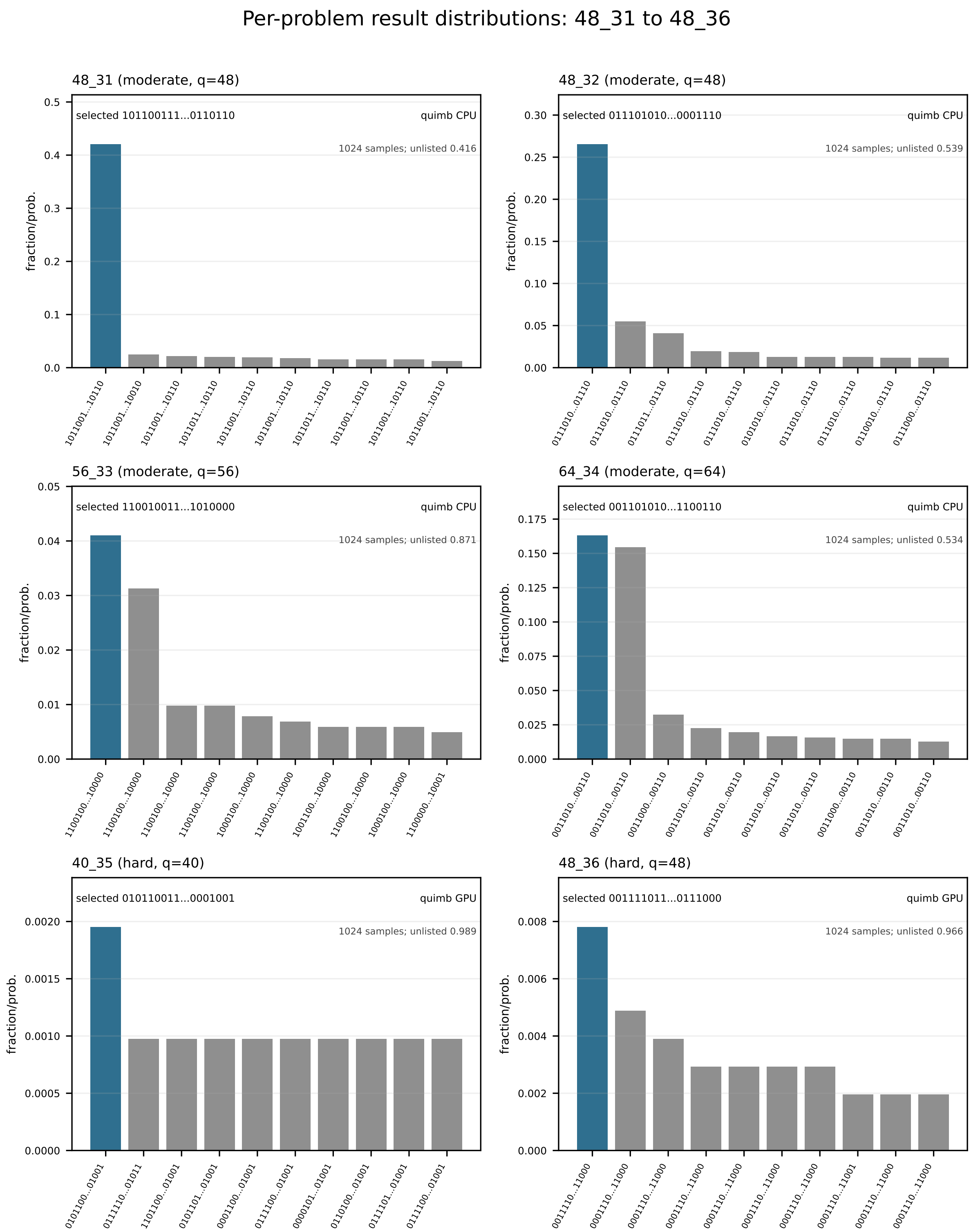

Figure 12: Per-problem result distributions, page 6. Bars show the top reported exact probabilities or sample fractions. The selected candidate is highlighted in blue when it appears among the displayed top outcomes; grey bars are alternate outcomes. Unlisted probability mass or sample fraction is reported inside each panel.

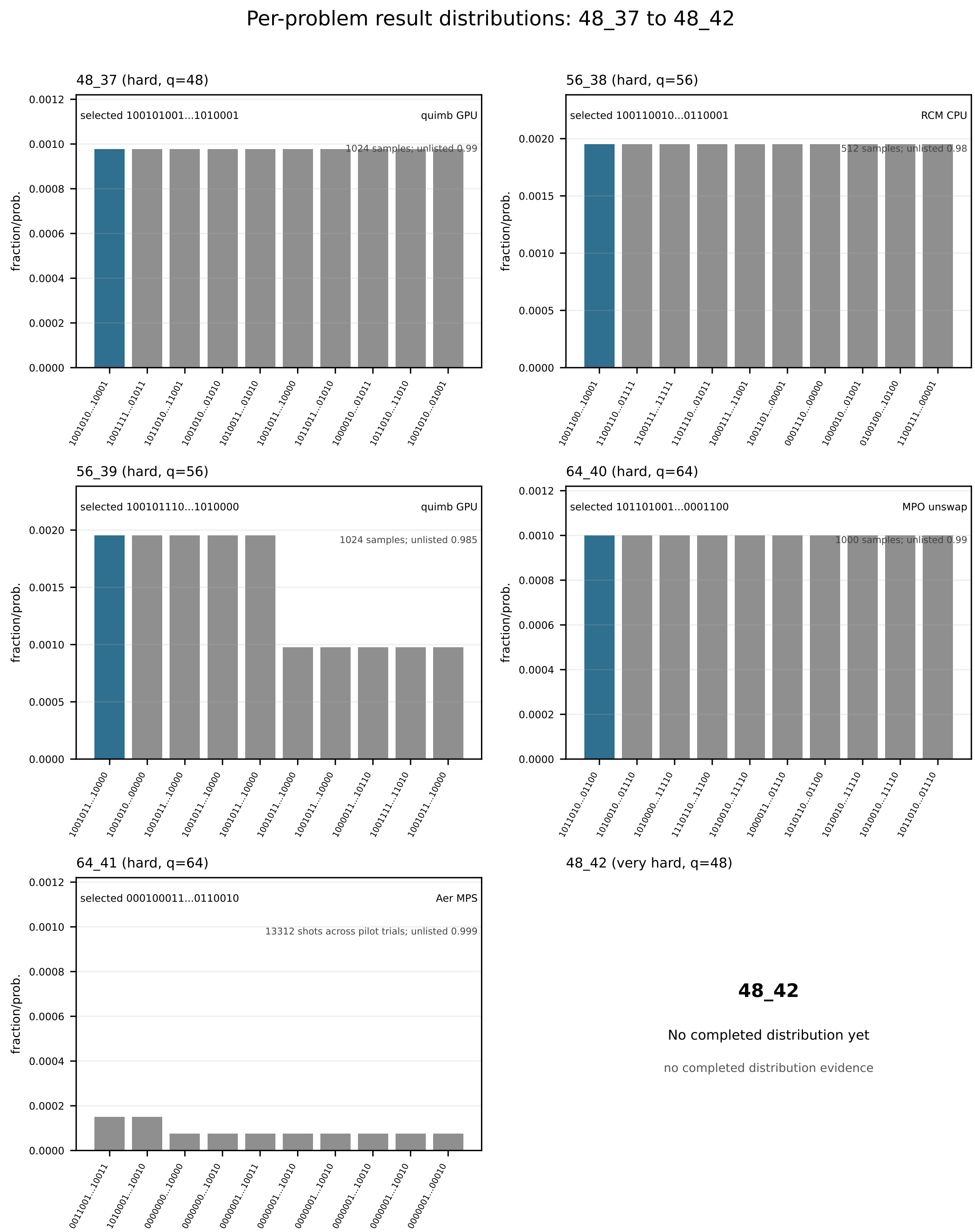

Figure 13: Per-problem result distributions, page 7. Bars show the top reported exact probabilities or sample fractions. The selected candidate is highlighted in blue when it appears among the displayed top outcomes; grey bars are alternate outcomes. Unlisted probability mass or sample fraction is reported inside each panel.

Per-problem result distributions: 56_43 to 96_48

56_43 (very hard, q=56)

56_43

No completed distribution yet

no completed distribution evidence

64_44 (very hard, q=64)

64_44

No completed distribution yet

no completed distribution evidence

72_45 (very hard, q=72)

72_45

No completed distribution yet

no completed distribution evidence

80_46 (very hard, q=80)

80_46

No completed distribution yet

no completed distribution evidence

88_47 (very hard, q=88)

88_47

No completed distribution yet

no completed distribution evidence

96_48 (very hard, q=96)

96_48

No completed distribution yet

no completed distribution evidence

Figure 14: Per-problem result distributions, page 8. Bars show the top reported exact probabilities or sample fractions. The selected candidate is highlighted in blue when it appears among the displayed top outcomes; grey bars are alternate outcomes. Unlisted probability mass or sample fraction is reported inside each panel.

Per-problem result distributions: 104_49 to 104_49

104_49 (very hard, q=104)

104_49

No completed distribution yet

no completed distribution evidence

Figure 15: Per-problem result distributions, page 9. Bars show the top reported exact probabilities or sample fractions. The selected candidate is highlighted in blue when it appears among the displayed top outcomes; grey bars are alternate outcomes. Unlisted probability mass or sample fraction is reported inside each panel.